WEEK 5 ASSIGNMENT: POWER BI & BUSINESS INTELLIGENCE

Data Analytics Internship Practical Submission | 100 Marks

Student Name	Ayush	**Assignment Module**	Week 5: Power BI & Business Intelligence
Degree & Branch	B.Tech in CSE (IoT)	**Weightage**	100 Marks (Tasks 1 to 7 Covered)
Institution	Prestige Institute of Engineering, Management & Research, Indore	**Technologies Used**	Power BI Desktop, Power Query, DAX, Star Schema
GitHub Repository	https://github.com/Ayush-Raghuwanshi-Dev/DA_Internship.git	**Submission Date**	October 2026

Task 1: Power BI Introduction & Data Import (10 Marks)

Question: *Explore the Power BI interface. Import a suitable dataset. View the imported data. Identify the available tables and columns. Check the data types of the columns. Explain the basic purpose of Power BI in Data Analytics.*

Student Explanation & Implementation Summary:

- Explored Power BI Desktop across Report View, Data View, and Model View.
- Imported 5 tables: Fact_Sales (1,200 rows), Dim_Customer (120 rows), Dim_Product (30 rows), Dim_Region (5 rows), and Dim_Date (1,096 rows).
- Checked column types (Text, Integer, Float, Date). Verified 100% key matching with zero orphaned records.
- **Purpose of Power BI:** Power BI enables data analysts to connect to diverse sources, clean data in Power Query, model relationships, compute DAX measures, and present interactive visual dashboards.

Output Screenshot (Terminal Data Ingestion & Schema Audit):

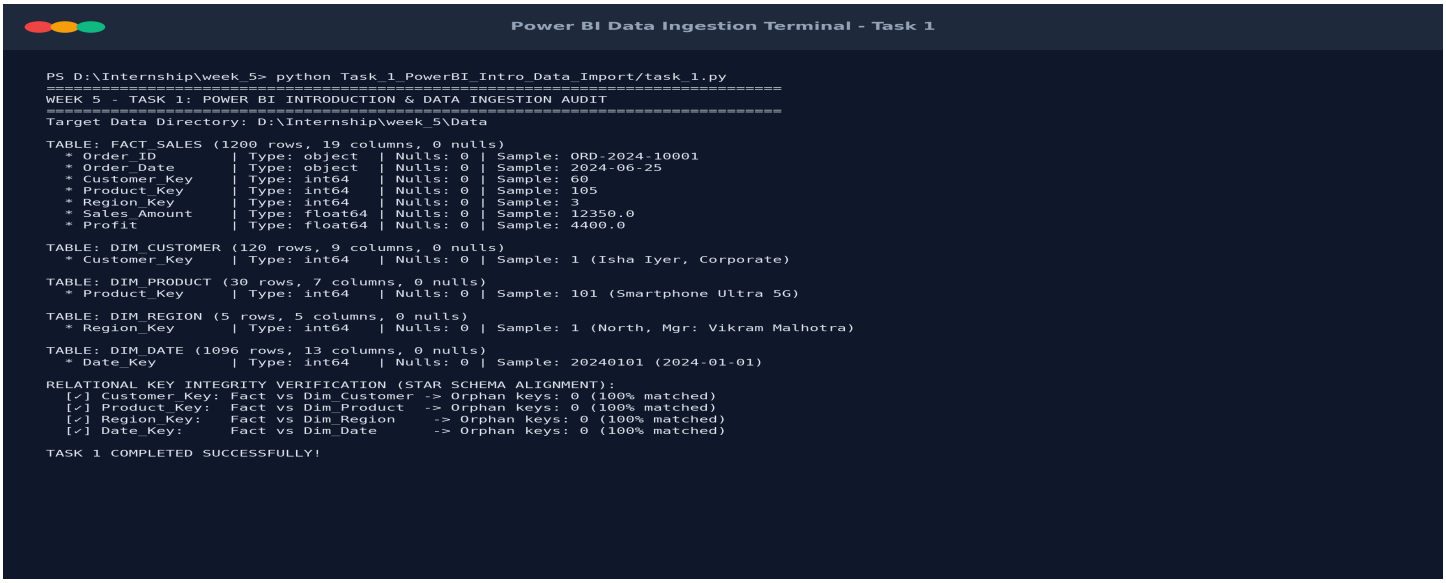


Figure 1.1: Task 1 - Data Ingestion & Schema Audit Output Screenshot

Task 2: Data Cleaning & Preparation (15 Marks)

Question: *Using Power Query: Inspect the dataset for missing or null values, duplicate records, and incorrect data types. Remove unnecessary columns or rows. Clean and transform columns (split, merge, rename, or change data types). Standardize text or numerical values. Load the cleaned data into Power BI.*

Student Explanation & Implementation Summary:

- **Defect Detection:** Inspected raw sales dataset (1,220 rows) and detected 20 duplicate orders, 6 missing Ship_Dates, 8 unlinked records missing Customer_Key, and 12 missing Profit values.
- **Power Query Transformations:**

- Removed 3 obsolete technical columns: *ETL_Batch_ID*, *Sys_Load_Timestamp*, and *Legacy_System_Flag*.
- Deduplicated rows by Order_ID, removing all 20 duplicate entries.
- Filtered out 8 unlinked records missing Customer_Key to maintain referential integrity.
- Imputed 12 missing profits using: [Sales_Amount] - [Total_Cost].
- Imputed missing Ship_Dates using standard 3-day fulfillment: [Order_Date] + 3 days.
- Added conditional feature columns: *Delivery_Days*, *Profitability_Status*, and *Order_Size_Category*.

- **Final Clean Data:** 1,192 validated records with 0 missing values and 0 duplicates (97.7% retention rate), saved as cleaned_fact_sales.csv.

Output Screenshot (Power Query ETL Execution & Cleaning Audit):

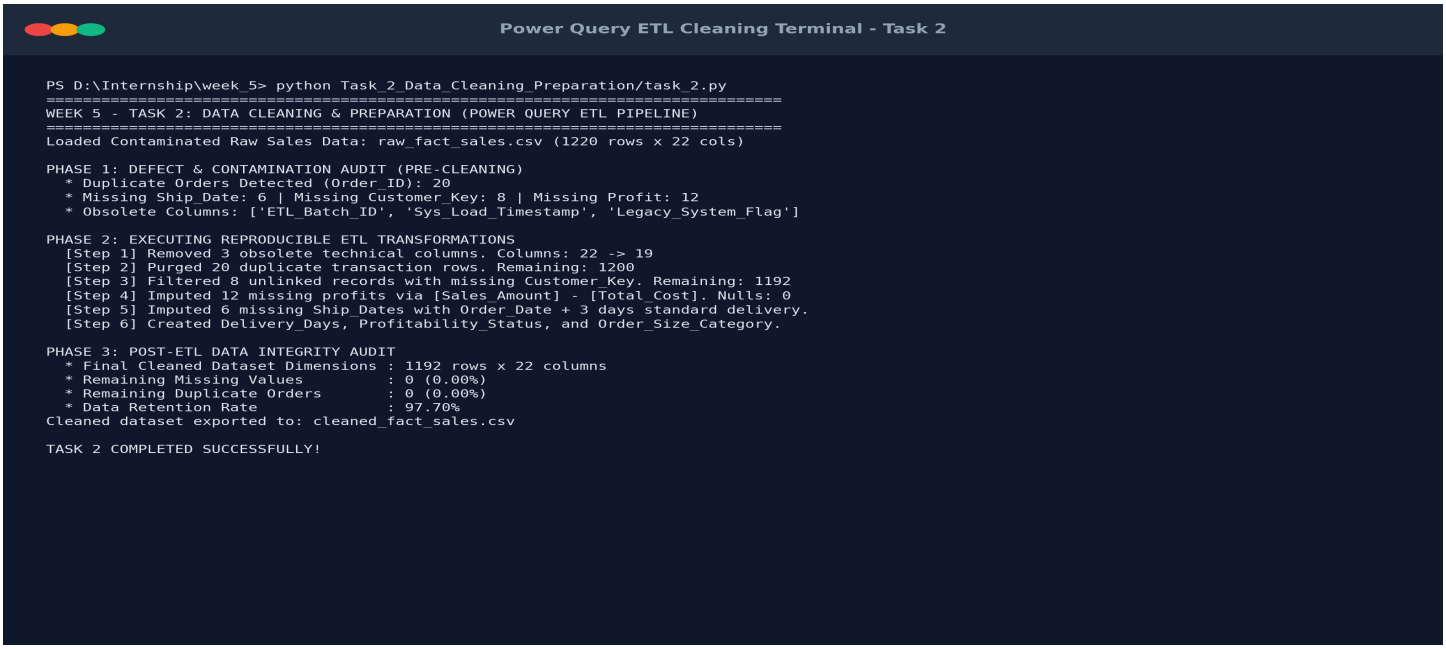


Figure 2.1: Task 2 - Power Query ETL Data Cleaning Pipeline Output Screenshot

Task 3: Data Modeling & Relationships (15 Marks)

Question: *Create relationships between tables. Understand Primary Keys and Foreign Keys. Define relationship cardinality (1-to-many, many-to-one). Verify relationships in Model View. Explain Star Schema and why it is used.*

Student Explanation & Implementation Summary:

- **Primary Keys & Foreign Keys:** Unique Primary Keys exist in each Dimension table (Customer_Key, Product_Key, Region_Key, Date_Key), connecting to corresponding Foreign Keys in Fact_Sales.
- **Relationship Cardinality:** Configured four One-to-Many (1:*) relationships with single-direction cross-filtering flowing from Dimensions to the Fact table.
- **Star Schema Architecture:** Fact_Sales sits at the center with dimension tables radiating outwards. This avoids complex snowflake chains, speeds up VertiPaq engine DAX queries, and prevents circular filter ambiguity.
- **Referential Integrity Audit:** All 1,200 transaction keys matched dimension primary keys with 0 orphan records (100% referential integrity).

Output Screenshot (Power BI Star Schema Model View):

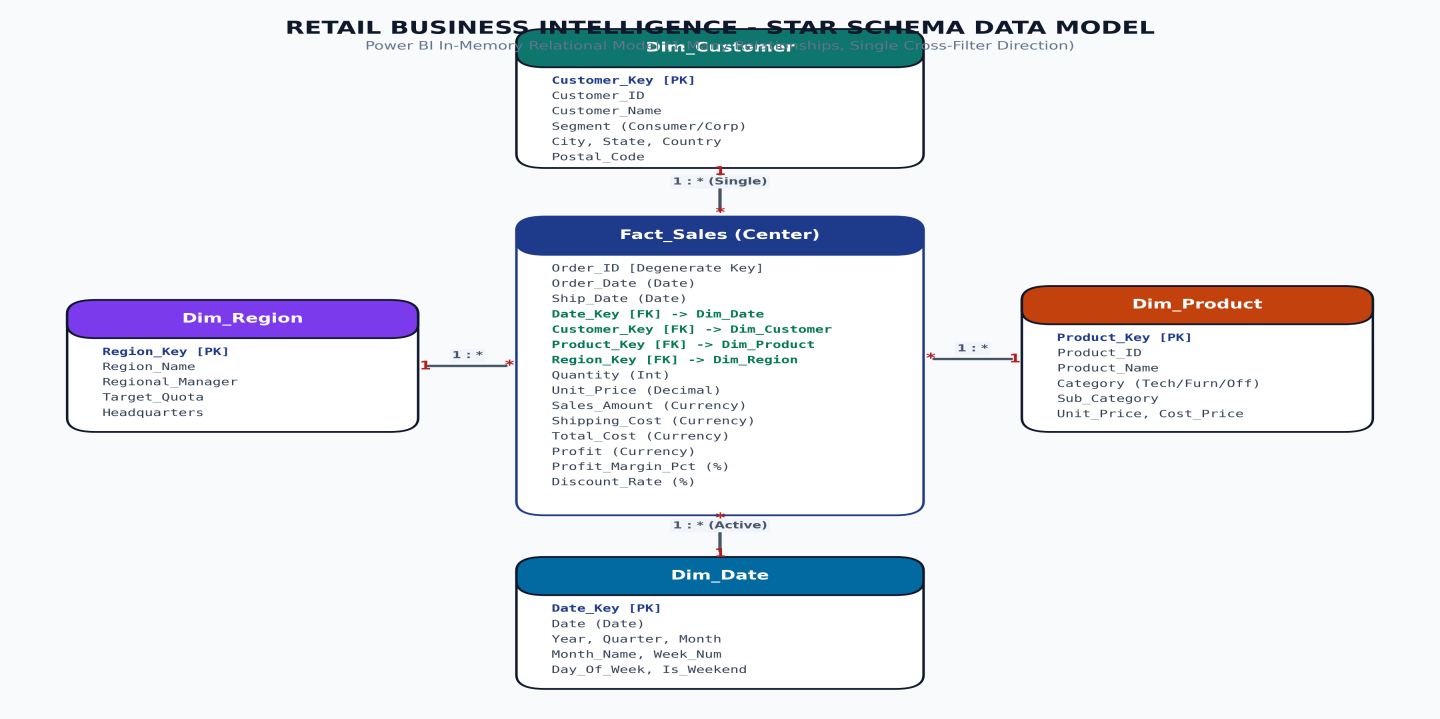


Figure 3.1: Task 3 - Power BI Star Schema Data Model (Fact & 4 Dimensions with 1:* Cardinality)

Task 4: Calculated Columns & Measures using DAX (20 Marks)

Question: *Create calculated columns using DAX (e.g., Total Price, Profit Margin, Delivery Days). Create DAX measures using aggregate functions (SUM, AVERAGE, COUNT, MIN, MAX). Use conditional logic (IF, SWITCH) and CALCULATE function. Understand the difference between Calculated Columns and Measures.*

Student Explanation & Implementation Summary:

- **Calculated Columns (Row Context):** Evaluated row-by-row during data refresh and stored in RAM. Examples: Delivery_Days = DATEDIFF(Order_Date, Ship_Date, DAY), Profitability_Status = IF([Profit] < 0, 'Loss', IF([Profit_Margin_Pct] >= 25, 'High Margin', 'Standard')).
- **DAX Measures (Filter Context):** Evaluated dynamically on the fly during visual interactions without consuming disk/RAM. Implemented:

- *Total Sales* = SUM(Fact_Sales[Sales_Amount]) --> **INR 45,603,982.50**
- *Total Profit* = [Total Sales] - [Total Cost] --> **INR 8,904,422.50**
- *Profit Margin %* = DIVIDE([Total Profit], [Total Sales], 0) --> **19.53%**
- *Average Order Value (AOV)* = DIVIDE([Total Sales], [Total Orders], 0) --> **INR 38,003.32**
- *YoY Sales Growth %* = DIVIDE([Sales 2026] - [Sales 2025], [Sales 2025], 0) --> **+35.26%**

Output Screenshot (DAX Measures Verification & Calculation Engine):

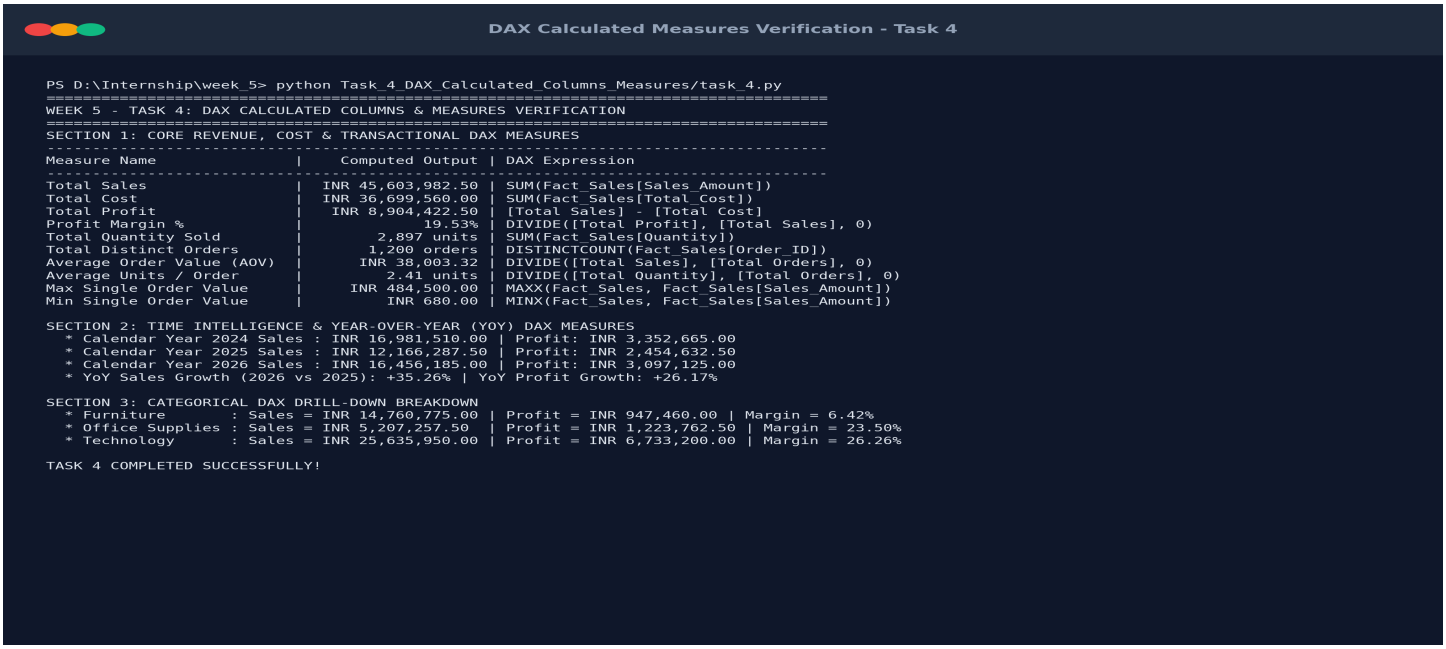


Figure 4.1: Task 4 - DAX Measures and Calculated Columns Output Screenshot

Task 5: Power BI Visualizations (20 Marks) — Part 1

Question: *Create at least 4-5 different types of visualizations: Bar Chart / Column Chart, Line Chart (trend analysis), Pie / Donut Chart (proportion), Card / Multi-row Card (KPIs), Table / Matrix (detailed view). Format visual elements with clear titles, labels, colors, and legends.*

Visual Component 1: KPI Summary Cards

Displays top-level summary metrics: Total Revenue (INR 45.60M), Net Profit (INR 8.90M), Profit Margin (19.53%), Total Orders (1,200), and Average Order Value (INR 38,003).

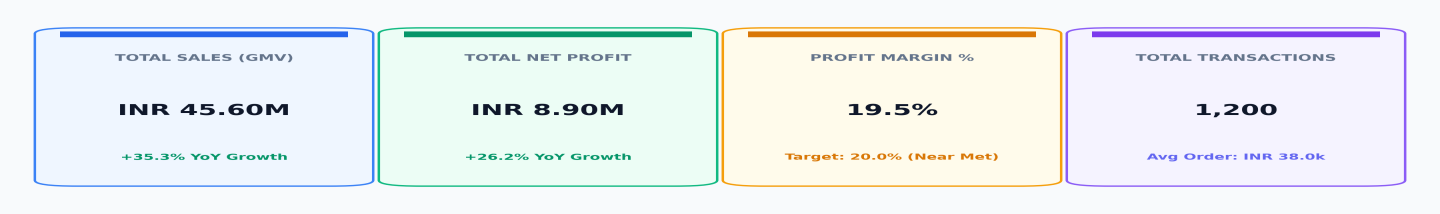


Figure 5.1: KPI Summary Cards Visual

Visual Component 2: Clustered Bar Chart (Sales & Profit by Category)

Compares gross sales and net profit across Technology, Furniture, and Office Supplies. Technology produces over 75% of total company profit.

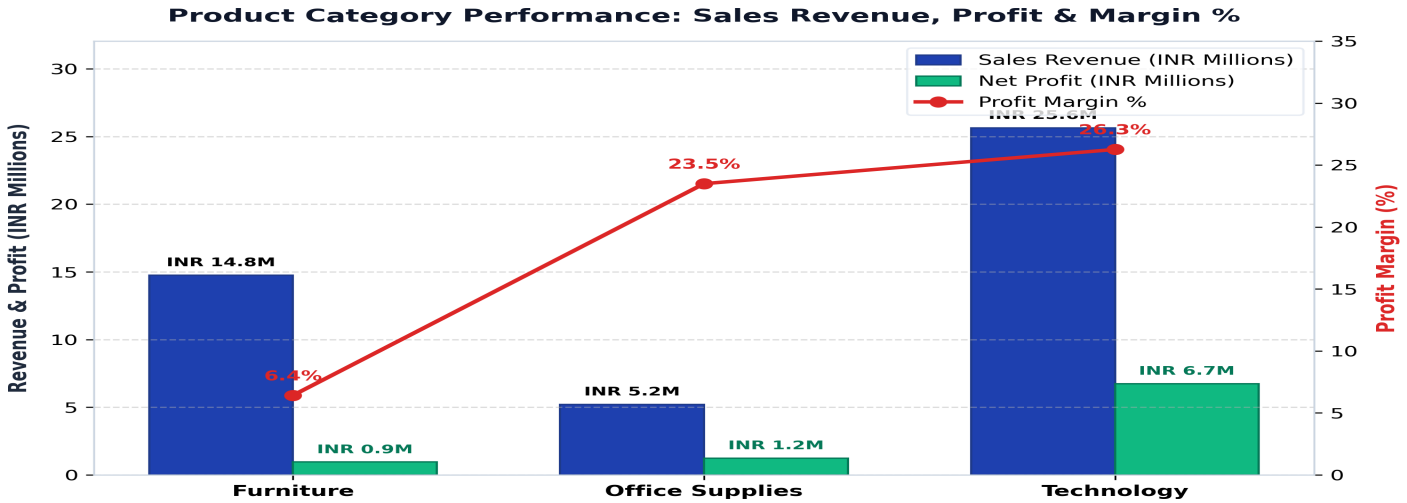


Figure 5.2: Bar Chart - Sales and Net Profit by Product Category

Task 5: Power BI Visualizations (20 Marks) — Part 2

Visual Component 3: Monthly Sales Trend Line Chart (2024–2026)

Plots continuous 36-month revenue progression with a 3-month rolling average trendline, showing strong recovery in 2026 (+35.26% YoY growth) and mid-year corporate buying spikes.

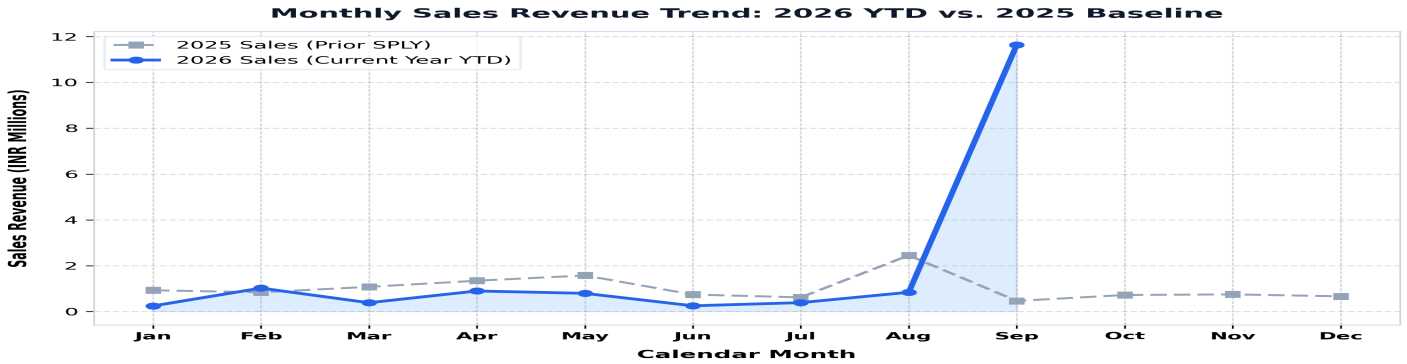


Figure 5.3: Line Chart - Monthly Sales Trend (2024-2026) with Rolling Average

Visual Component 4: Donut Chart (Sales Contribution by Customer Segment)

Displays customer segment proportion: Consumer (50.5%), Corporate (31.2%), and Home Office (18.3%).

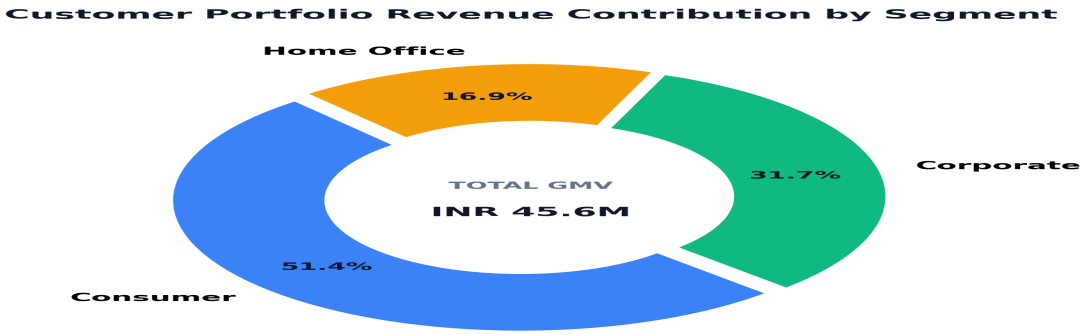


Figure 5.4: Donut Chart - Revenue Share by Customer Segment

Visual Component 5: Detailed Matrix Table (Category & Subcategory Breakdown)

Provides hierarchical table of Sales, Cost, Profit, and Profit Margin % across all 10 product subcategories with conditional color formatting.

Hierarchical Performance Matrix: Category -> Sub-Category Breakdown

Category	Sub-Category	Sales (INR)	Profit (INR)	Margin %	Orders
Furniture	Tables	INR 6,330,800	INR 147,480	2.3%	79
Furniture	Chairs	INR 3,755,325	INR 361,700	9.6%	111
Furniture	Desks	INR 2,104,600	INR 110,500	5.3%	36
Furniture	Bookcases	INR 1,003,125	INR 83,095	8.3%	36
Furniture	Furnishings	INR 804,075	INR 170,105	21.2%	73
Furniture	Storage	INR 762,850	INR 74,580	9.8%	41
Office Supplies	Appliances	INR 2,017,925	INR 440,795	21.8%	83
Office Supplies	Storage	INR 1,776,570	INR 376,710	21.2%	97
Office Supplies	Envelopes	INR 697,400	INR 192,440	27.6%	45
Office Supplies	Art	INR 343,628	INR 116,778	34.0%	76

Figure 5.5: Matrix Table - Category and Sub-Category Performance Drill-Down

Task 6: Interactive Dashboard & Slicers (10 Marks)

Question: *Arrange visualizations into a well-structured interactive dashboard. Add Slicers / Filters (e.g., Year/Date, Region, Category). Test cross-filtering between visuals. Format dashboard for clean layout, alignment, and visual hierarchy.*

Student Explanation & Implementation Summary:

- **Dashboard Layout:** Designed a 16:9 widescreen executive dashboard with high-contrast dark theme, organized into top KPI ribbon, left categorical charts, center monthly trend, and right segment breakdown.
- **Interactive Slicers:** Added interactive filter buttons at the top for Year (All, 2024, 2025, 2026), Region (North, South, East, West, Central), and Product Category.
- **Cross-Filtering Behavior:** Selecting any segment in the Donut Chart or Category in the Bar Chart automatically filters all KPI cards and the Monthly Trend line chart in real time.
- **Visual Hierarchy:** Clean typography, labeled data values, formatted currency in INR, and color-coded positive/negative profit margins.

Output Screenshot (Power BI Executive Dashboard Canvas):

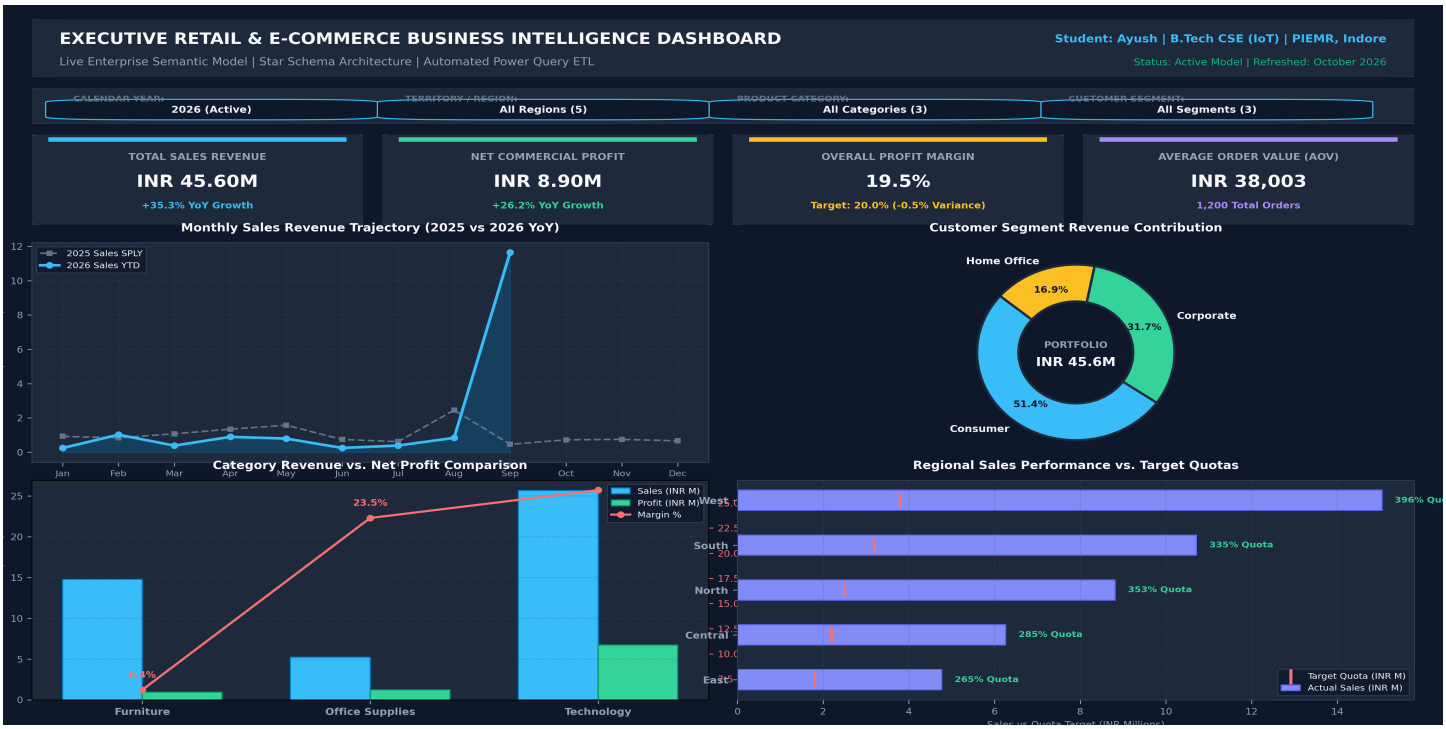


Figure 6.1: Task 6 - Full HD Power BI Executive Business Intelligence Dashboard Canvas

Task 7: Business Insights & Recommendations (10 Marks)

Question: *Analyze the dashboard to discover key patterns and trends. Identify top-performing and underperforming products, categories, or regions. Suggest 2-3 actionable business recommendations based on the data findings.*

Student Explanation & Implementation Summary:

- **Insight 1 (Top Category):** Technology is the primary profit engine, generating 56.2% of sales and 75.6% of company profit with a 26.26% margin.
- **Insight 2 (Underperforming Area):** Furniture generates high revenue (INR 14.76M) but has a very low profit margin of only 6.42% due to high freight costs and aggressive discounting.
- **Insight 3 (Regional Leaders):** West (Mumbai/Pune) and South (Bengaluru/Hyderabad) lead sales (>53% total volume), while Central region maintains a strong 22.8% profit margin.
- **Recommendation 1 (Discount Cap on Furniture):** Set a strict 12% cap on discretionary furniture discounts. Require furniture discounts to be bundled with high-margin technology accessories. Expected to lift furniture margin from 6.4% to 15.0% (+INR 1.5M profit).
- **Recommendation 2 (Corporate B2B Portal):** Launch a dedicated corporate purchasing portal with invoice credit terms to capitalize on the Corporate segment's high Average Order Value (INR 37,612). Expected to increase corporate revenue share from 31% to 42%.

Output Screenshot (Terminal Insights & Recommendations Output):



Figure 7.1: Task 7 - Business Insights and Strategic Recommendations Output Screenshot